# PTS

**Pan / Tilt / Slider**

A five-mount motorised camera system for live video production - wireless, self-healing, and controllable from everything in the booth.

PTS drives five motorised camera mounts over a dedicated wireless network. Each mount pairs a Teensy 4.1 motion controller (TMC2209 silent stepper drivers, StallGuard homing) with an ESP32 bridge and round touchscreen; a central hub links every mount to the operator's control surfaces. Positions, speed presets and subject calibrations live on the mounts themselves - controllers can come and go mid-show.

## System architecture

**Control surfaces**
PC app (PyQt6, joystick, CV tracking) · web app on phones · 7" hub touchscreen · Bitfocus Companion + QLab via OSC

- - - - LAN / WiFi / USB - - - -

**HUB - ESP32-S3**
WiFi AP 'CamMount' · TCP / WebSocket / OSC servers · USB serial to PC · UART to hub display · autonomous self-recovery

- - - - ESP-NOW wireless star (touch-screen pairing, channel self-healing) - - - -

| CAM 1 | CAM 2 | CAM 3 | CAM 4 | CAM 5 |
|---|---|---|---|---|
| ESP32 AMOLED bridge | ESP32 AMOLED bridge | ESP32 AMOLED bridge | ESP32 AMOLED bridge | ESP32 AMOLED bridge |
| Teensy 4.1 + TMC2209 | Teensy 4.1 + TMC2209 | Teensy 4.1 + TMC2209 | Teensy 4.1 + TMC2209 | Teensy 4.1 + TMC2209 |
| pan · tilt · slider · zoom | pan · tilt · slider · zoom | pan · tilt · slider · zoom | pan · tilt · slider · zoom | pan · tilt · slider · zoom |

## Highlights

| Feature | What it does |
|---|---|
| **Look-at tracking** | Camera keeps aiming at a calibrated 3D subject while the slider travels - subjects are calibrated in two taps and switchable mid-move from any controller. |
| **Zero-config pairing** | No MAC addresses or per-unit firmware: hold a mount's screen, pick CAM 1-5, tap your hub. The hub learns mounts on first contact; conflicts resolve with one tap on the hub display. |
| **Self-healing** | Autonomous recovery ladders on hub and mounts (reinit then restart), idle maintenance restarts, watchdogs at every layer, and a motion-side dead-man: a mount stops within 500 ms of losing its control stream. |
| **Health telemetry** | Every node reports heap, loop timing, link quality and reset cause every 10 s into the PC log - degradation is visible hours before it becomes a symptom. |
| **Show-control ready** | OSC servers on both the hub and the PC app put every function on Stream Deck buttons and QLab cue stacks (reference overleaf). |

# OSC control reference - Companion / QLab

Two OSC servers listen on **UDP port 9700** and accept the same commands, so Companion pages fire unchanged against either target. **Feedback is sent by the hub only -** point Companion at the hub for buttons that light up to show what the rig is doing:

| Target | Address | When to use |
|---|---|---|
| **The hub itself** | `192.168.4.1` | No PC needed - hub + display + mounts + Stream Deck is a complete rig. Reach it via a WiFi-to-LAN bridge joined to the CamMount AP, or any machine joined to CamMount directly. |
| **PC app machine** | `that PC's IP` | When the PC app is running anyway. Configurable via osc_enabled / osc_port in the app config. |

Companion: add a **Generic: OSC** connection with the target IP and port 9700, then use its Send message actions as below. Integer arguments preferred; floats (QLab network cues) are accepted.

## Command reference

| Address | Arguments | Action |
|---|---|---|
| `/pts/estop` | - | E-STOP every mount |
| `/pts/cam/N/estop` | - | E-STOP mount N |
| `/pts/cam/N/goto` | `slot 1-10` | Recall a stored position (uses the mount's active speed presets) |
| `/pts/cam/N/store` | `slot 1-10` | Store the current position |
| `/pts/cam/N/clear` | `slot 1-10` | Clear a stored position |
| `/pts/cam/N/jog` | `pan tilt slider zoom` | Velocities -1000..1000; re-streamed at 20 Hz until stopped |
| `/pts/cam/N/jog/stop` | - | Stop jogging (put on the button's release action) |
| `/pts/cam/N/jog/left /right /up /down` | `1 press, 0 release` | Pan and tilt from plain buttons; full deflection at the active preset speed |
| `/pts/cam/N/jog/slide/left /slide/right` | `1 / 0` | Slide along the rail |
| `/pts/cam/N/jog/zoom/in /zoom/out` | `1 / 0` | Zoom |
| `/pts/cam/N/speed/pt` | `1-4` | Active pan/tilt speed preset |
| `/pts/cam/N/speed/sl` | `1-4` | Active slider speed preset |
| `/pts/cam/N/speed/pt/up /down` | - | Pan/tilt preset one step, clamped at 1 and 4 |
| `/pts/cam/N/speed/sl/up /down` | - | Slider preset one step, clamped |
| `/pts/cam/N/speed/pt/inc /sl/inc` | - | Cycle the preset 1-2-3-4-1 |
| `/pts/cam/N/subject` | `0-7` | Select look-at subject; switches live during a move |
| `/pts/cam/N/lookat` | `0 or 1` | Look-at slider move to min (0) or max (1) with the selected subject |
| `/pts/refresh` | - | Resend all feedback for every mount |
| `/pts/cam/N/refresh` | - | Resend all feedback for mount N |

N = camera 1-5. Slots and speed presets are 1-based, subjects 0-based - matching every screen in the system.

## Feedback - buttons that show the rig

The hub sends state back to whoever last commanded it, so a Stream Deck shows which shot a camera is on, whether a mount is online and what speed it is set to - with nothing tracked in Companion. Enable **Listen for Feedback** on the OSC connection, then add an *OSC: Listen for OSC messages (Integer)* feedback matching an address and value below.

| Address | Value | Meaning |
|---|---|---|
| `/pts/cam/N/active` | `0 / 1` | Mount is online |
| `/pts/cam/N/state` | `int` | Mount state; 0 is idle |
| `/pts/cam/N/target` | `0-10` | Slot being moved to, 0 = not moving to one |
| `/pts/cam/N/speed/pt` | `1-4` | Active pan/tilt preset |
| `/pts/cam/N/speed/sl` | `1-4` | Active slider preset |
| `/pts/cam/N/slot/M/state` | `0-3` | 0 empty, 1 occupied, 2 moving here, 3 arrived |

Slot state is one address per slot carrying one value, so a shot button colours itself from a single feedback with nothing to combine - the same stored / moving-to / arrived picture the hub display and the web app show. Only changes are sent, so a rig at rest is silent; everything is resent every 5 s, and in full the first time an address speaks to the hub, so a surface that joins late catches up on its own. **If nothing arrives**, check Companion's Source Port is the port it also listens on, and that its machine is on the same subnet as the hub - commands still arrive across a subnet mismatch, replies cannot get back, and UDP reports no error either way.

### Button recipes

| Button | Setup |
|---|---|
| **Recall shot 3, cam 2** | Press: `/pts/cam/2/goto` with int 3 |
| **Hold-to-jog pan-left, cam 1** | Press: `/pts/cam/1/jog/left` int 1   Release: same address, int 0 |
| **Shot button that shows its state** | Press: `/pts/cam/1/goto` int 7. Feedback on `/pts/cam/1/slot/7/state`: 1 red (stored), 2 amber (on its way), 3 green (arrived). |
| **One speed button per camera** | `/pts/cam/1/speed/pt/inc` cycles 1-2-3-4-1 per press. Add a feedback on `/pts/cam/1/speed/pt` to show which preset is live. |
| **Look-at pair, cam 3** | Button A: `/pts/cam/3/subject` int 2   Button B: `/pts/cam/3/lookat` int 1 (press A mid-move to retarget live) |
| **QLab cue** | Network cue to port 9700: `/pts/cam/2/goto 4` - camera moves fire from the cue stack, in time with light and sound. |
| **Show-stopper** | A big red `/pts/estop` (no argument). |

### Safety

**-** OSC jogs are re-streamed at 20 Hz with a 15 s TTL, so a lost release message cannot run a mount away; prefer goto / lookat for long moves.
**-** The mounts' own 500 ms dead-man is the final backstop: if the hub or PC dies mid-jog, motion stops regardless of the controller.
**-** E-STOP over OSC clears held jog streams before broadcasting the stop.